Henry Hsu

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SUMMARY

Software Engineering Intern at Google and former GenAI Team Lead, shipping agentic AI workflows in production: memory, tool APIs, MCP, and hybrid graph/vector retrieval. Grew a B2B platform's users from 3K to 20K while cutting LLM token usage by 67%. LlamaIndex contributor with 14 merged PRs. Now pursuing an M.C.S. at Rice University.

WORK EXPERIENCE

Software Engineering Intern — Jun. 2026 – Aug. 2026

Google — *Taipei, Taiwan*

- **Integration Test Infrastructure:** Built **3** test infrastructures from a **zero** baseline (Java, Python), onboarding **400+** integration test suites across **200+** smart home device types, with **90%+** unit test coverage for the infrastructure itself
- **Agentic Failure Triage:** Developed an LLM agent that triages scheduled integration test runs and surfaces root-cause summaries directly on issue tickets, replacing manual log inspection and cutting failure narrow-down time by **20+ minutes** per issue
- **Local Device Reproduction:** Extended the agent to reproduce device failures locally against virtual devices instead of waiting on cloud runs, cutting manual reproduction time by **85%**; **shipped and adopted by the team**

Generative AI Team Lead — Dec. 2024 – Aug. 2025

MaiAgent Co., Ltd. *– Award-Winning B2B GenAI Startup* — *Taipei, Taiwan*

- **Building AI Agents:** Upgraded chatbots into AI agents with memory systems, tool APIs, and **Model Context Protocol (MCP)** clients, driving **120%** partner growth (CTBC Bank, MSI, HPE, iGroup), **3K** to **20K** user growth, and **67%+** lower LLM token usage
- **Contributing to Open Source:** Contributed **14** merged pull requests to the **LlamaIndex** open-source project (**15K+ stars**), fixing bugs and adding features in integrations with AWS Bedrock, Claude, Elasticsearch, Cohere, OpenAI, MCP Client, and AI Agent Workflow
- **CI/CD & Testing:** Built a **GitHub Actions** CI pipeline with pytest unit and E2E coverage, reaching **67%** from a **zero** baseline; cut production hotfixes by **90%** initially and **50%** long term
- **Technical Roadmap & Leadership:** Defined GenAI product roadmap for a 10-person startup; led end-to-end delivery of production features including Agentic RAG, Artifact Generation, Tagging & Permission System, and Information Retrieval

Generative AI Intern — Sep. 2024 – Dec. 2024

MaiAgent Co., Ltd. *– Award-Winning B2B GenAI Startup* — *Taipei, Taiwan*

- **Pipeline Scaling:** Refactored with an **asyncio**/**Celery** async framework and dual-database synchronization (RDB/Vector DB), achieving **3.5x** faster parsing and scaling from **3M** to **20M+** text chunks while **doubling** indexing performance
- **Reranker Integration:** Integrated Cohere and BGE open-source rerankers as a customer-selectable retrieval feature, improving RAG precision on large documents by **30% Precision@5** and **7% Precision@10**

EDUCATION

Rice University - Top 20 U.S. University — Aug. 2025 – Dec. 2026 (Expected)

M.C.S., Computer Science (GPA: 4.00/4.00) — *Houston, TX*

National Yang Ming Chiao Tung University (NYCU) - Top 3 Taiwan University — Sep. 2020 – Jun. 2024

B.S., Industrial Engineering and Management (GPA: 4.07/4.30, Two-Time Dean's List) — *Hsinchu, Taiwan*

Minor: Computer Science (Domain GPA: 4.13/4.30)

PROJECTS

Agentic Hybrid RAG | *Graph RAG (Neo4j, Cypher), Vector RAG (Milvus), Multi-Agent (LangGraph)* — **Feb. 2024 – Aug. 2024**

- **Self-Routing Agentic RAG:** Built a query-classification agent that routes each request to **Graph RAG**, vector RAG, or a hybrid path, picking global or local retrieval by question type
- **ML Clustering:** Applied K-means and DBSCAN over the vector store to select pivotal nodes for a low-cost Graph RAG, saving **35%** of indexing token cost for only 2.56% worse RAGAS scores
- **Hybrid Retrieval:** Fed Graph RAG results into vector retrieval with **HyDE**, reaching **88.2%** MAR@10 on multi-hop datasets

AWARDS

Presidential Hackathon Winner (2024, Top 5 Nationally) - Urban noise agent: structured-output LLM pipeline and tool calling

23rd Golden Peak Award (2025) - Outstanding Commercial Product, MaiAgent AI Platform

SKILLS

Languages: Python, Java, C/C++, TypeScript | **GenAI:** LlamaIndex, LangChain, LangGraph, MCP | **Fullstack:** Django, FastAPI, React
Databases: PostgreSQL, MongoDB, Milvus, Neo4j, Elasticsearch | **Tools:** git, Docker, Shell Script, Nginx
Data Science: numpy, pandas, scikit-learn, PyTorch, JAX | **AWS:** Bedrock, EMR, EC2, S3, RDS, ElastiCache, SageMaker